Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

% Facts sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

```
11 ?- consult(['c:/Users/Lenovo/Downloads/exp7.1.pl']).
true.

12 ?- sentence(X).
X = the_cat_sits .
```

Exercise 2: Compound Term Unification

% Facts likes(john, food(pizza,

spicy)). likes(mary, food(pizza,

sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

```
13 ?- consult(['c:/Users/Lenovo/Downloads/exp7.2.pl']).
true.

14 ?- likes(john, food(pizza, T)).
T = spicy .
```

Exercise 3: Conflict Resolution

% Facts rule1(climate_change).

rule2(climate_change).

% Rule conflict(X) :- rule1(X),

rule2(X).

% Query

% ?- conflict(climate_change).

Output:

```
15 ?- consult(['c:/Users/Lenovo/Downloads/exp7.3.pl']).
true.

16 ?- conflict(climate_change).
true.
```

Result

Thus, the Unification and Resolution exercises were successfully executed in PROLOG, and the expected results were obtained.